

Zi3n took the paper out of my hands as soon as I finished copying the two grids of numbers. “I already know how to do it. That’s too easy,” he said. To my surprise, he got it right. The week after having passed all his subjects with good marks, he decided he wanted to learn how to multiply matrices. Zi3n is just a 4th grader living in a shelter (Hogar Ntra. Sra. de los Desamparados) after having migrated from the Philippines, so he surely didn’t understand linear algebra, but was genuinely interested in the strange math that I was teaching him. That day, I internalised something that isn’t written in textbooks but defines my mindset: curiosity, initiative, and discipline can overcome any background. It’s not about the opportunities you encounter, but the decisions you take that make a meaningful impact.

I was never one of the smart kids at school. I was more like the average kid until one day I decided to put some work and discipline into actually studying what I was taking, so that my marks went up. And sure enough, they did indeed. It took at first a lot of time because I simply wasn’t used to retaining all information and understanding stuff (after all, I kind of just went through middle school on autopilot), but I managed to hit the GPA ceiling of 10/10 with multiple honour mentions in grade 10.

Furthermore, I conducted some research, especially in grade 11. In grade ten, I researched on my own about [Einstein’s 1905 theory of special relativity](#), a research project tailored towards an audience not familiar with the topic. In grade eleven, I doubled down with research and produced a research project about the [bosonic string theory](#) and a research paper (unpublished, in Spanish) about [experimental energy sources and trends in energy sources](#).

As my research projects grew more complex, I realised there wasn’t any good markup language to let me express ideas efficiently, so I made [CSScribe](#), my custom markup language for producing research papers and projects. I managed to create this custom markup language because I’m all self-taught in Computer Science (never taken a single class). I completed Harvard’s CS50 (2023 recording, online, no official certificate, wasn’t willing to pay \$220 for it) and Soy Dalto’s

courses on HTML, CSS, JavaScript and Python. I taught myself with official documentation, React, TailwindCSS and bits of Rust.

However, what honestly drives me and makes me happy is helping people in some kind of way. I do think that's true for every human being. Zión is one of the stories I have. I "tutored" this kid for around seven months or so. It was not only about tutoring, but also offering some emotional support and spending time together, since many of these kids did not have a family. I tutored him on whatever he needed help with (he failed Natural Science, and then passed that same exam), played soccer with a paper ball on the basement floor, and we did just about anything. He was kind of my son sometimes. I do miss volunteering there (places are full this year).

In grade ten, I met Allenger, who spoke Spanish but had never received a single class about Spanish syntax or grammar. He came from South America. I tutored him for five months, around two hours per week, usually. He finally managed to pass the Spanish subject in grade 10, graduate, and pursue vocational training in graphic design.

Alberto de la Torre was another guy I tutored, this one a bit more intensive. Grade 11 corresponds to *1º de bachillerato* in Spain, when marks start becoming really important. Alberto was taking STEM, which was a nightmare for him since he really struggled with all science-related subjects, and he wanted help (even though he was already receiving private tutoring from outside). I voluntarily helped him for a couple of hours per day the last few days of grade 11 when I already had no exams, and he managed to hit a 7.4 GPA after having had 5.6 just three months earlier.

Grade 11 was a very special year for me because I randomly woke up one day and decided to found the school's Programming Club. I don't know where that idea came from. I joked about it with a friend for two days, and just a few weeks later, I opened the door to the IT room, and there were 30 people there. In reality, it wasn't that simple, but it literally felt like it. I just went with my friend through grades 8 to 11 on a Monday with a poster saying the usual stuff you can say about a club, and on Friday, we had the IT room overcrowded (there was even a guy from grade 12, no idea

how he found out). I'd like to say it was a full Programming Club, but it honestly was more like a social club with a couple of laptops involved sometimes. [Abscissa](#) was born there, I developed [CSScribe](#) as well, and we removed spy software our school had downloaded into students' laptops (some switched to Linux directly).

I have also faced some challenges, most of them being sought, honestly. The biggest one, and one I'm still dealing with, is having left my old high school (Colegio Mater Immaculata). This was just one month ago, and it hit me hard because 80% of my social life was wiped out overnight. There were multiple facts that made me and my family take this decision, but long story short, I don't really need classes to learn the same stuff, and scholarships were simply not coming this year. I formally have a new (public) high school, but it is not a physical one; there are no lessons, and I just take the exams. I have to study everything on my own, which is fine for me, though I do miss the social life I had previously (which took me 14 years to build), and I'm now rebuilding it outside high school with more volunteering.

This month, I took the GED (General Education Development) to get a US high school equivalency. I am going to complete my education in Spain, and get my national high school diploma, but at the start of this month, I was feeling like taking it. To be clear, the GED is not instead of my national high school diploma; it is in addition to. I took all the GED subjects in 19 days, so technically I'm already graduated by US standards, I guess, and scores do look pretty nice.

Another challenge I found for myself was [CSScribe](#), as I mentioned earlier. There are simply no good tools to produce research projects and papers. You can play around with Microsoft Word, which is a toy when you need extensive math, formatting, or simply Mermaid diagrams and code blocks; or you can go all in on LaTeX with Overleaf, which has a really steep learning curve and is simply not a good option for small projects. So I made [CSScribe](#) to solve that (from CSS, and Scribe, which means to write in Latin). I haven't updated it in a couple of months, mainly because it

needs a new full parser and removing a third-party dependency for compiling to PDF, but I will come back to it. It has a [Visual Studio Code extension](#) with 40+ downloads.

I'm someone who can adapt quickly, so thinking long-term is hard for me. Not because I don't, but because I can't predict the future. If a year ago someone had told me all I've accomplished in these last 12 months, I wouldn't have believed it. So all I have is a vague idea of what I'm going to do; I simply can't say more. Here's my best attempt at communicating what I'm going to do in the future with my life, though:

[Abscissa](#). A platform for solving math problems interactively and sharing them. No more multiple choice questions, no more fill-in-the-gaps. That's literally my marketing pitch (I did some light marketing on X, just to get some beta-testers). I usually also compare it with LeetCode, which is a platform for learning how to code. Right now, it is very, very green and underdeveloped, but I got the main piece of the puzzle, a way of typing math into computers that's natural for humans and computable for machines. Abscissa is powered by [Mathsys](#), which is a markup-programming language hybrid that is both compilable to LaTeX for mathematical typesetting on-screen and parseable so that the [Mathsys](#) runtime actually does verify the math. Honestly, explaining it confuses people for some reason, so it can be tried online on the [Abscissa playground](#) (though only LaTeX compilation is available, no computing yet, otherwise it would overwhelm my server). I will continue working on [Abscissa](#) and [Mathsys](#) in the future. Abscissa already has 80+ users, and made (funny story) a total revenue of 0.05€.

Research. Since I left my old high school, no one is really pushing me to actually conduct research, but I'm doing it anyway. I only plan to produce one research paper this year, because I'm going to publish it on arXiv, and who knows if somewhere else as well. This is going to be my first proper published research (I don't really count my research on [my GitHub profile](#) as published, very few people would take something published there seriously), and I hope that by late December, it's already finished. I have indeed already started it, and I won't say much, but it's about literal energy

(electricity) as a currency, from a theoretical point of view. Physically, it does make sense, because energy is capable of work by definition, and it is protected against inflation, deflation and speculation by the laws of thermodynamics (we really don't want to break those). However, in practice, it is much harder, and will kill centralised economies along with the concept of a "global economy" since energy losses over distance would push towards self-reliance and independence.

Skills. Knowledge is cheap these days. With the Internet, it seems like successful people emerge from anywhere; be it Finland (Linus Torvalds), South Africa (Elon Musk), the UK (Tim Berners-Lee), or anywhere else. They all share that they didn't just push knowledge frontiers, but they implemented it as well, and the implementation was so useful that it pushed knowledge and capabilities (not before!). After all, it makes a lot of sense. What's actually valuable is stuff that solves problems, but knowledge does not always solve problems. Skills that implement knowledge are, of course, always going to be more valuable than just knowledge alone. All this to say, I'm focusing and will continue to focus on skills like marketing, coding, and problem-solving, without disregarding physical skills like woodworking, driving, cooking, or other skills that improve day-to-day life or provide value to the world without a massive time investment before delivery.

I do not plan to return to Spain, but I do not wish to stay in the US to live after college either. Location choice is a really personal choice, so I'm drawn to make the decision that makes me happier long term. For me, I think happiness comes from being social and having a sense of purpose and belonging. In the future, I'll probably settle in a slower-paced environment in which genuine relationships can flourish naturally. My current guesses are around rural Argentina, Paraguay and Albania; whilst I make an impact on the world through entrepreneurship in IT or other STEM fields.